

STORM LAKE, IA, USA

WMO: 725496

Lat: 42.597N

Lon: 95.241W

Elev: 1488

StdP: 13.92

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-3.9	-16.8	2.3	-7.9	-11.3	3.2	-3.8	34.3	15.7	31.8	17.3	11.5	290	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	91.0	73.7	89.5	73.4	86.1	72.0	78.5	86.4	76.7	84.8	74.9	83.0	13.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	146.3	84.1	73.4	131.1	81.6	72.3	126.6	80.7	43.2	86.1	41.3	84.6	39.4	83.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	26.4	23.8	-16.4	96.4	6.1	2.8	-20.9	98.4	-24.4	100.0	-27.9	101.5	-32.3	103.5
			-16.5	81.2	7.5	1.9	-21.9	82.6	-26.3	83.7	-30.5	84.7	-36.0	86.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	20.1	23.0	35.7	48.8	60.8	70.9	73.9	70.8	64.5	51.3	36.9	24.0
	DBStd	21.64	13.10	12.55	12.63	10.52	9.25	6.72	5.83	5.43	8.35	9.85	10.99	11.61
	HDD50	3637	926	757	469	144	17	0	0	0	4	104	410	806
	HDD65	6864	1391	1176	909	492	194	23	6	12	110	437	843	1271
	CDD50	3100	0	1	25	109	351	626	742	646	439	143	18	0
	CDD65	853	0	0	1	7	63	200	283	193	95	11	0	0
	CDH74	7871	0	0	12	137	703	1937	2625	1444	871	137	5	0
	CDH80	2674	0	0	1	40	244	722	954	405	274	34	0	0
Wind	WSAvg	11.4	12.7	12.7	13.0	14.2	13.2	10.8	8.2	7.4	8.9	11.2	12.3	12.3
Precipitation	PrecAvg	32.0	0.5	0.6	1.7	3.3	3.9	4.9	4.3	4.4	3.4	2.2	1.4	0.7
	PrecMax	45.4	3.5	2.9	5.5	11.6	7.8	11.9	11.5	10.3	12.4	8.4	4.2	4.3
	PrecMin	13.9	0.0	0.0	0.0	0.2	0.8	0.5	0.6	0.2	0.7	0.3	0.0	0.0
	PrecStd	7.8	0.6	0.6	1.4	2.4	1.6	2.8	2.5	2.4	2.4	1.7	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	58.7	74.9	85.6	91.4	94.7	95.1	91.1	91.2	85.6	71.8	54.1
		MCWB	42.6	49.6	57.9	62.1	68.1	73.8	77.4	76.7	71.2	64.3	56.2	47.1
	2%	DB	45.0	50.0	67.5	77.4	86.1	90.6	90.7	88.0	86.5	77.4	63.5	47.6
		MCWB	39.2	44.7	54.8	58.6	65.5	71.9	76.4	76.5	70.4	61.2	52.5	42.5
	5%	DB	39.1	43.5	60.5	72.1	81.6	87.9	88.3	84.2	82.2	72.3	57.1	42.6
		MCWB	35.4	39.1	51.9	56.0	63.1	70.0	75.9	74.7	68.3	60.2	47.7	37.6
	10%	DB	36.2	38.7	54.2	66.2	77.0	84.1	85.8	81.5	79.3	67.6	52.3	37.4
		MCWB	33.9	35.5	47.3	52.9	61.3	68.6	74.8	72.2	66.3	56.8	45.6	34.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	50.5	61.7	64.7	72.5	79.0	81.7	80.4	76.8	70.6	59.2	49.8
		MCDB	52.2	56.6	71.6	75.1	84.7	88.0	89.5	87.7	83.8	78.9	66.6	51.7
	2%	WB	39.2	44.7	56.6	61.1	69.0	76.0	79.2	77.7	73.8	65.5	53.8	41.8
		MCDB	44.1	49.6	63.5	73.4	80.4	85.1	87.0	85.2	81.7	74.5	62.4	46.9
	5%	WB	36.1	39.3	51.9	58.0	66.5	73.8	77.3	75.5	71.8	61.5	49.7	38.4
		MCDB	38.9	43.7	60.0	68.9	77.1	83.1	85.6	82.5	79.7	68.6	56.1	42.0
	10%	WB	33.6	35.8	46.4	54.8	63.9	71.5	75.4	73.6	69.5	57.2	45.4	35.6
		MCDB	35.8	38.9	53.8	64.8	72.3	81.2	84.1	79.8	76.2	64.9	51.3	38.2
Mean Daily Temperature Range	5% DB	MDBR	15.6	16.0	17.9	21.0	20.6	19.9	18.7	17.9	22.1	20.2	18.2	14.8
		MCDBR	20.0	20.6	27.9	30.8	27.2	23.6	20.3	20.6	25.3	27.1	25.5	19.8
		MCWBR	15.7	15.7	16.7	15.8	11.8	10.1	11.2	12.1	12.3	14.3	15.6	15.2
	5% WB	MCDBR	18.3	19.5	26.1	25.9	22.6	19.7	18.4	17.7	20.5	22.4	23.5	17.6
		MCWBR	14.7	15.1	16.5	15.1	12.1	11.3	11.6	11.7	12.1	14.9	15.2	14.1
Clear-Sky Solar Irradiance	taub	0.278	0.290	0.316	0.365	0.390	0.406	0.416	0.407	0.370	0.330	0.291	0.277	
	taud	2.415	2.399	2.421	2.311	2.311	2.322	2.273	2.295	2.364	2.461	2.499	2.452	
	Ebn at Noon	276	291	295	286	280	275	271	270	272	272	270	267	
	Edh at Noon	24	29	32	39	40	40	42	39	34	27	22	22	
All-Sky Solar Radiation	RadAvg	605	939	1208	1521	1790	1986	2079	1747	1417	967	660	496	
	RadStd	57	72	110	134	129	160	111	121	90	120	42	36	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page